



**TED UNIVERSITY**

**SENG 491**

*Software Engineering*

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**SkillHood**

**Analysis Report**

**Kİİ**

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## 1. Introduction

This document is written for the Analysis and Specifications Report of SkillHood app. The complete requirements baseline is defined such that: what the system must do, how it should behave, and the constraints it must fit in. This report is the primary reference for the development team, course supervisor, and project jury throughout all phases.

In SkillHood, the app lets people share their knowledge without any financial transaction. All exchanges are made by an economy that rewards genuine, clear contribution. The report covers user management, live and service-based exchanges, skill discovery, a dynamic credit engine, reputation tracking, intelligent matching, and scheduling.

This document presents the analysis phase of the SkillHood project, including an overview of the proposed system, a comparison with existing solutions, functional and nonfunctional requirements, system models, and user interface mockups. The goal is to establish a clear and shared understanding of what SkillHood will do and how it will do it, forming a solid foundation for the design and implementation phases that follow

## 2. Current System

Currently, no dedicated platform exists that combines skill-based exchanges, a structured credit economy, scheduling tools, and a reputation system in a single application. Existing solutions either charge money for learning and services — excluding users who lack financial resources — or rely on informal community arrangements that have no accountability, no quality control, and no way to scale. Many platforms focus on only one side of the equation, such as freelancing marketplaces or online course providers, but fail to create a reciprocal environment where every user can be both a provider and a learner. As a result, people with valuable skills to share but limited budgets remain locked out, and the potential for community-driven skill exchange goes largely unrealized.

(Paraphrase yapmadım)

## 3. Proposed System

### 3.1 Overview

SkillHood is a platform for both web and mobile. A peer skill exchange app built on an inside credit economy. From requester to provider, every completed exchange triggers a credit transfer. It is calculated by a Credit Engine using:  $\text{BaseRate} \times \text{SkillValue} \times \text{DemandFactor} \times \text{ReputationFactor}$ . Each factor is related to platform configuration, skill taxonomy, live marketplace data, and the provider's rating history.

### 3.2 Functional Requirements

- **REQ-1:** System shall allow users
- **REQ-2:** System shall
- **REQ-3:** System shall provide a
- **REQ-4:** System shall

### 3.3 Nonfunctional Requirements

- **Performance:** Skill search results returned within 2 seconds; credit calculations completed within 1 second of confirmation.
- **Availability:** 99.5% uptime target; scheduled maintenance communicated at least 24 hours in advance.
- **Security:** TLS 1.3 for all data in transit; bcrypt password hashing; JWT-based authentication; GDPR and KVKK (Law No. 6698) compliance.
- **Usability:** Responsive layout across desktop and mobile; WCAG 2.1 Level AA conformance.
- **Scalability:** Backend designed for horizontal scaling; queries optimised for growth.
- **Maintainability:** Consistent naming conventions, inline documentation, versioned RESTful API documented via OpenAPI/Swagger.

### 3.4 Pseudo Requirements

- PR-1:**
- PR-2:**
- PR-3:**
- PR-4:**
- PR-5:**

## 3.5 System Models

### 3.5.1 Scenarios

#### Scenario 1: Live Language Session

- **Stimulus:** A user logs into SkillHood and searches for intermediate English practice. She filters by Language and Live Only, selects another user's listing (4.8 stars, 45 min, 25 credits, available Thursday at 6:00 PM), and submits a session request.
- **System Response:** SkillHood validates other user's availability and forwards the request with a push notification. Marco accepts; both users receive a confirmation alert and the time slot is reserved. After the session, both mark it completed. The Credit Engine applies the formula ( $\text{BaseRate} \times \text{SkillValue} \times \text{DemandFactor} \times \text{ReputationFactor}$ ), deducts 25 credits from Aisha, credits Marco's balance, and prompts both users to submit a rating.

#### Scenario 2: Service-Based Exchange – Garden Planting

- **Stimulus:** A user named Lena creates a service request describing a balcony garden planting task (approximately 2 hours, deadline Saturday morning) and posts it on the platform.
- **System Response:** SkillHood validates the request, publishes it to the Skill Discovery Feed, and notifies compatible providers. Caner accepts; the system sets the status to In Progress and notifies Lena. After Caner marks the task Delivered and Lena confirms completion, the system transfers 40 credits to Caner and prompts both users for a review.

#### Scenario 3: Dispute Resolution

- **Stimulus:** Melih completes a UI/UX portfolio review and marks the task Delivered. The requester disputes the delivery, claiming the feedback was incomplete.
- **System Response:** SkillHood marks the session as Disputed, freezes the 35 pending credits, and alerts platform moderators. The moderator reviews the exchange history and submitted feedback, determines Melih's delivery was substantive, and resolves the dispute in his favour. The frozen credits are released and transferred, and both parties receive a resolution notification.

### 3.5.2 Use Case Model

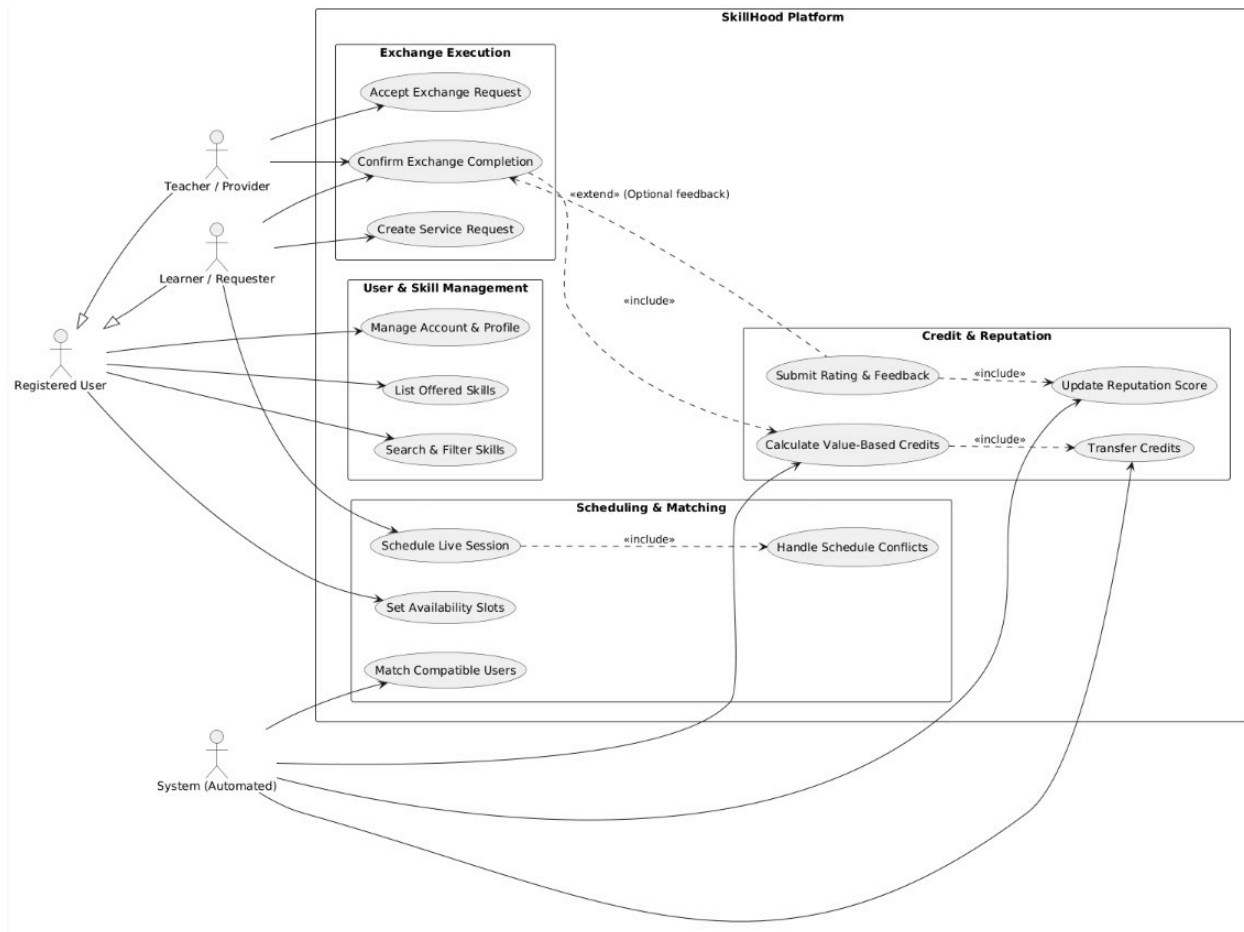


Figure 1 - Use Case Model

### 3.5.3 Object and Class Model

## Analysis Report for SkillHood

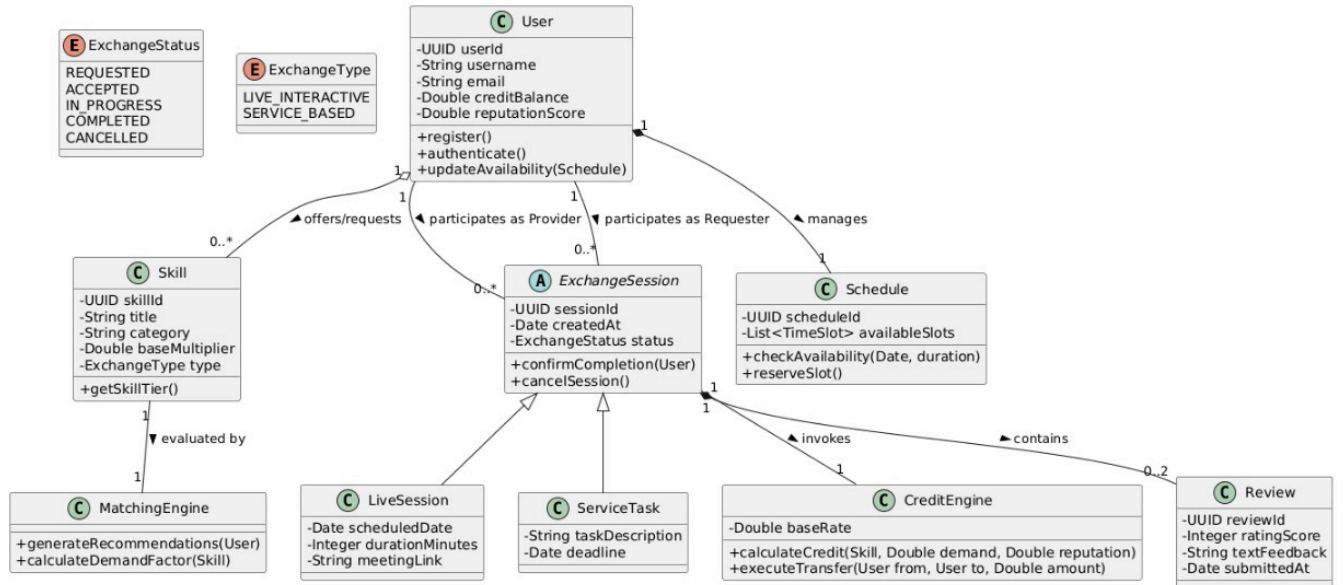


Figure 2 - Class Diagram

### 3.5.4 Dynamic Models

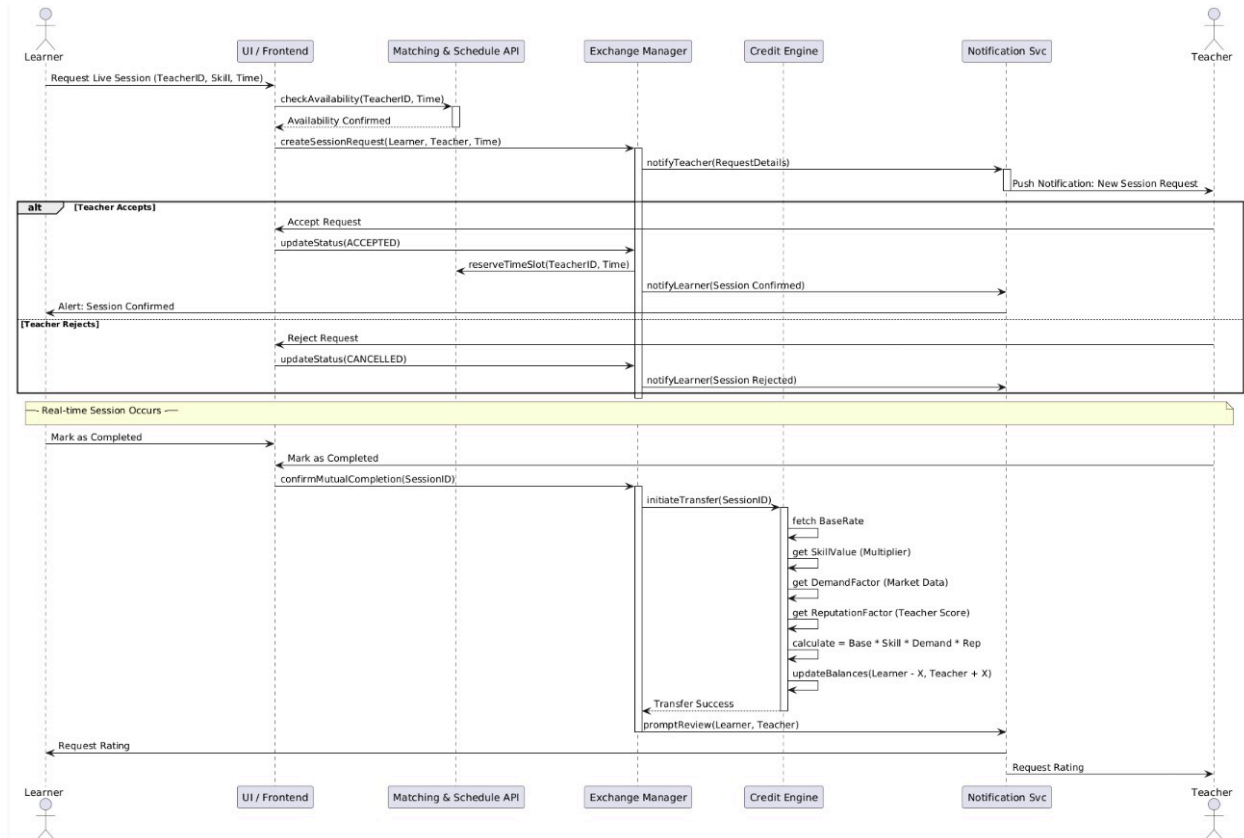


Figure 3 - Sequence Diagram

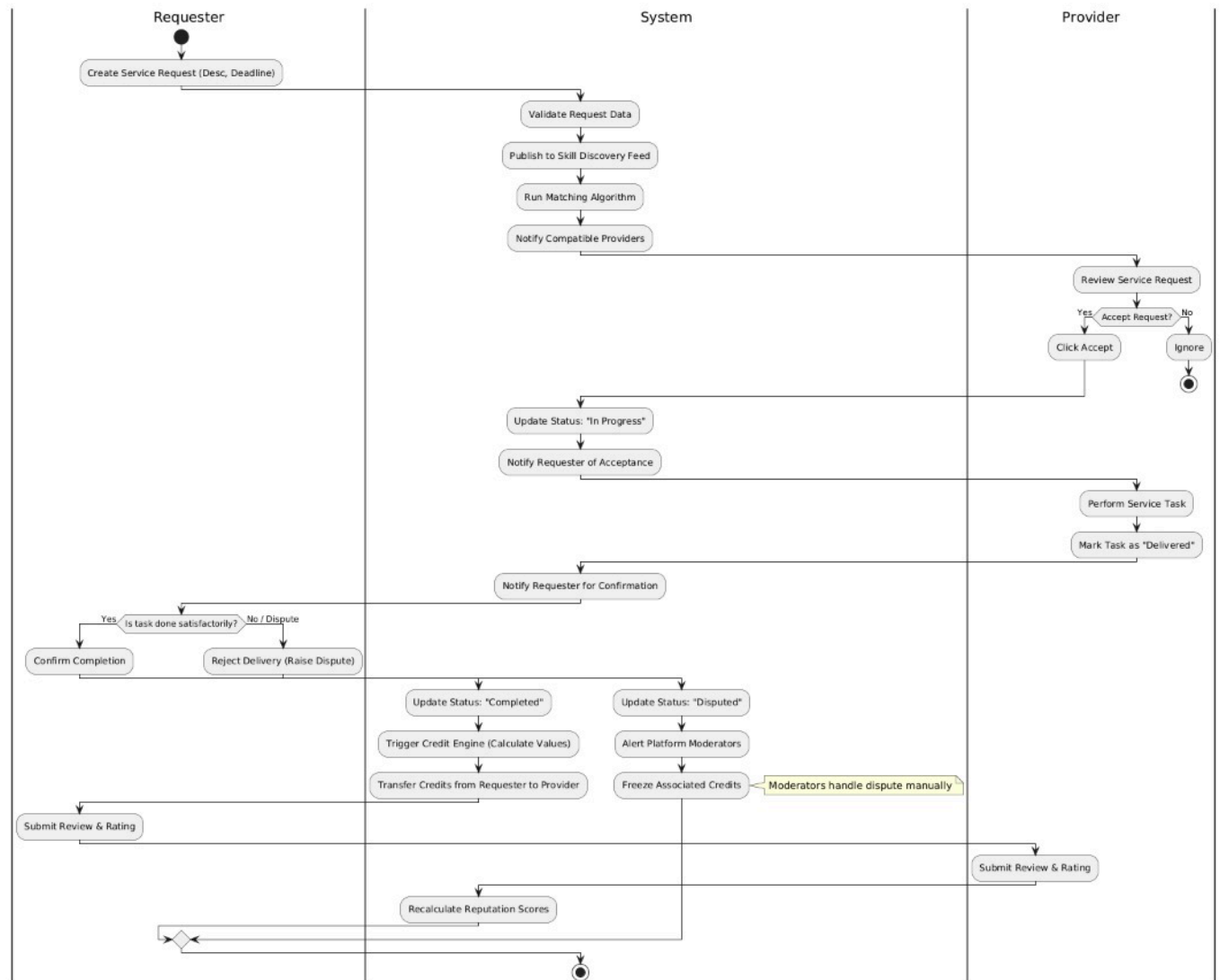


Figure 4 - Activity Diagram



### 3.5.5 User Interface – navigational paths and screen mock-ups

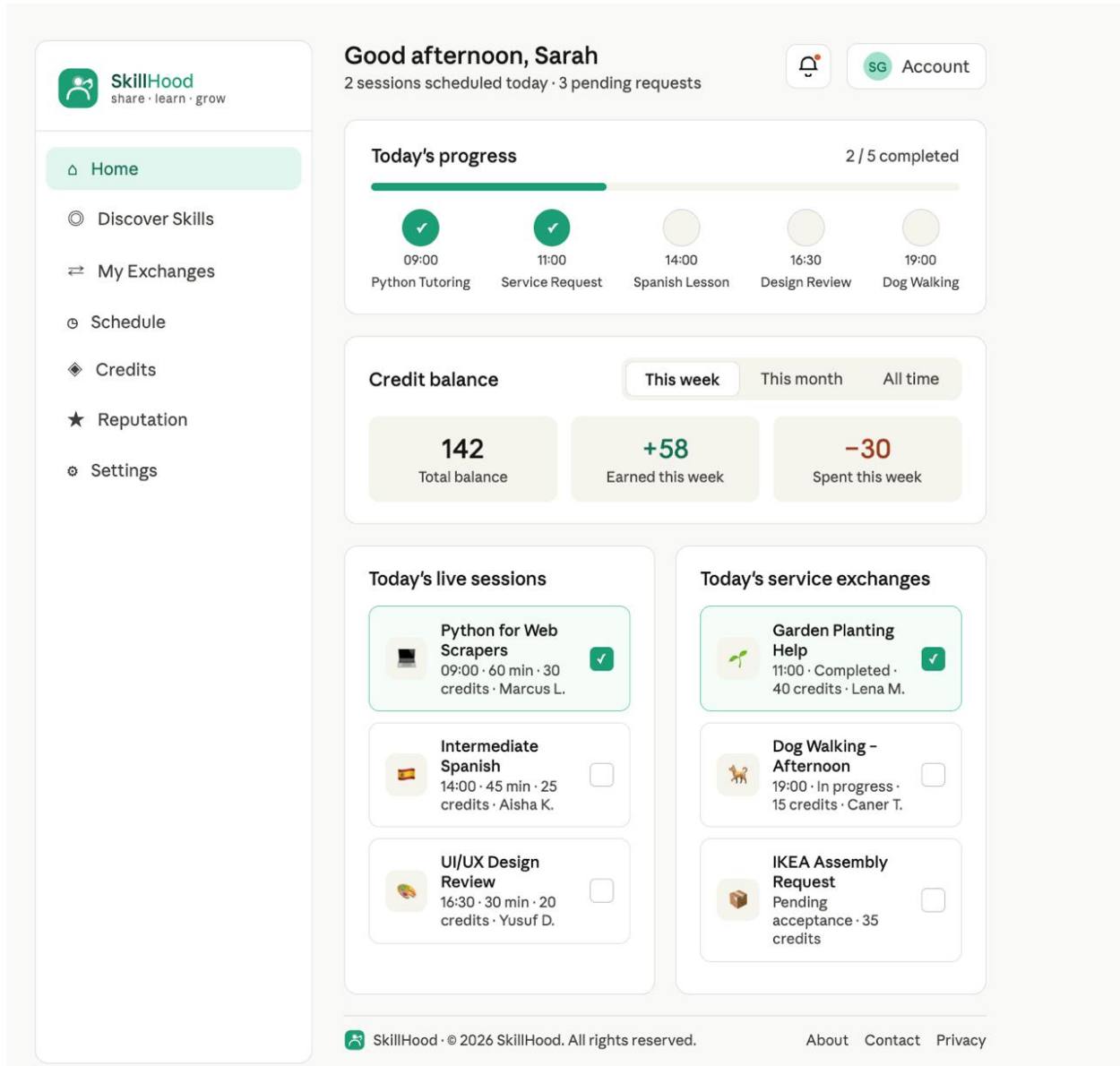


Figure 5 - Home Page Mock-up

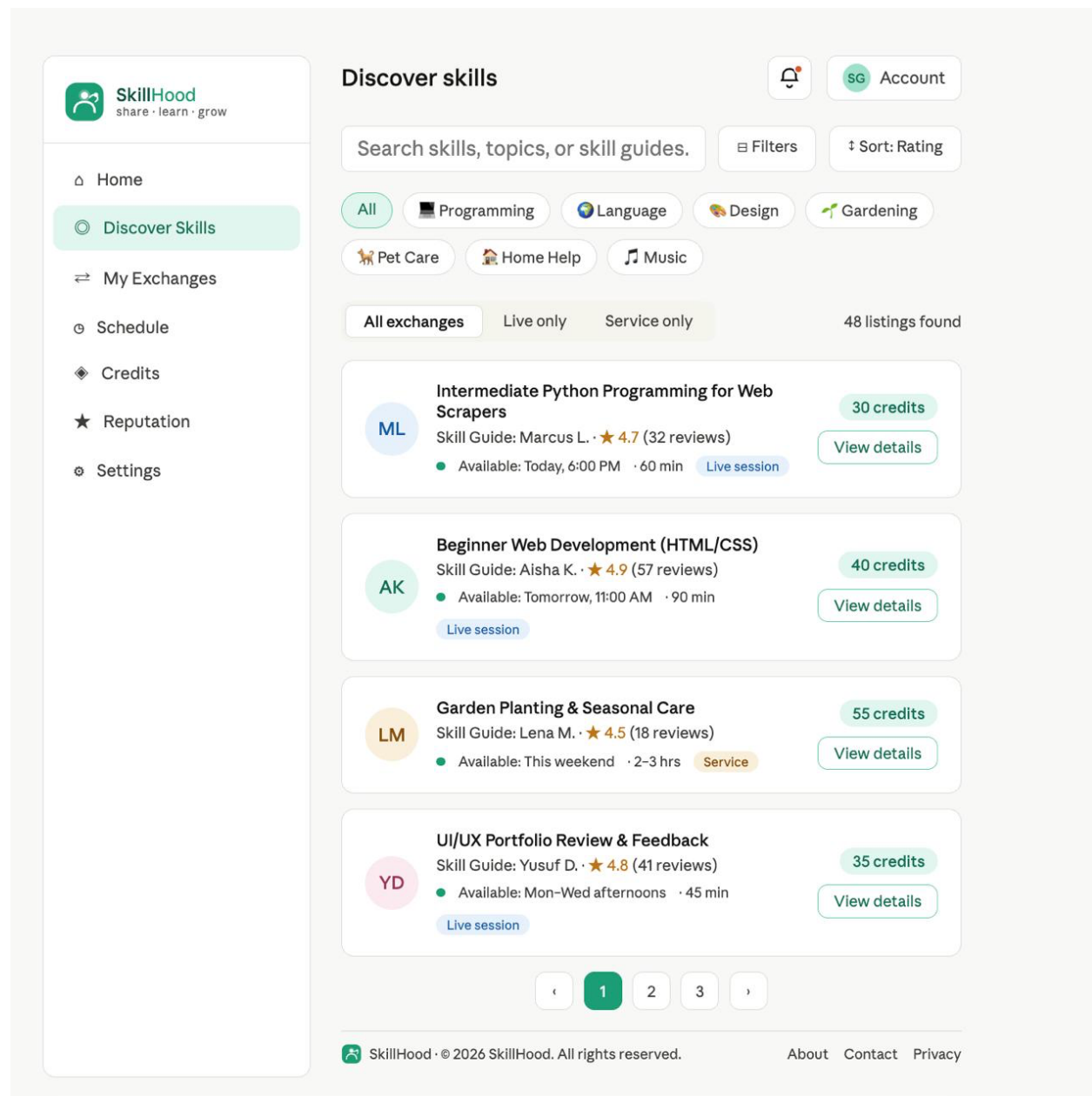


Figure 6 - Discover Skills Page

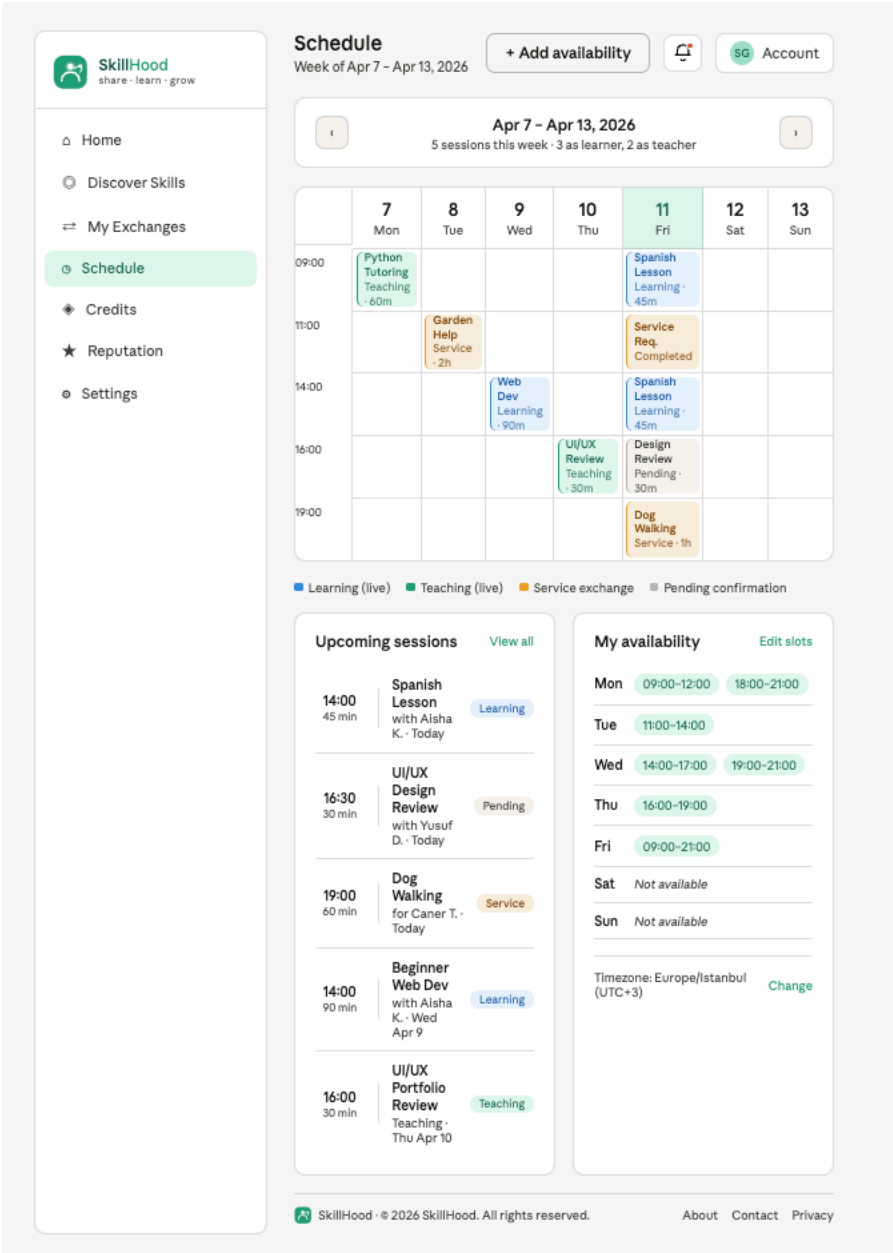


Figure 7 - Schedule Page

4.Glossary

|          |                                                           |
|----------|-----------------------------------------------------------|
| Stimulus | Input base rate, skill category, user rating, demand data |
|----------|-----------------------------------------------------------|

5. References

[1] IEEE Computer Society, "IEEE Recommended Practice for Software Requirements Specifications," IEEE Std 830-1998, 1998.

[2] B. Bruegge and A. H. Dutoit, "Object-Oriented Software Engineering, Using UML, Patterns, and Java," 2nd ed. Upper Saddle River, NJ, USA: Prentice-Hall, 2004.

[3] GeeksforGeeks, "What is Data Encryption?," Nov. 22, 2024. [Online]. Available: <https://www.geeksforgeeks.org/what-is-data-encryption/>

[4] M. Thoker, "UML Relations for Class Diagrams: Everything You Need to Know," Nov. 22, 2024. [Online]. Available: <https://thoker.medium.com/uml-relations-for-class-diagrams-everything-you-need-to-know-c784fa2645bb>

[5] Visual Paradigm, "UML Class Diagram Tutorial," Nov. 22, 2024. [Online]. Available: <https://www.visual-paradigm.com/guide/uml-unified-modeling-language/uml-class-diagram-tutorial/>